Exact Symplectic Flow and the Complete Stability–Resonance Atlas of the Ideal Penning Trap

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Abstract

We solve the ideal axially symmetric Penning Hamiltonian throughout its signed field–axial-frequency plane. One entire rotating-frame formula gives the exact six-dimensional symplectic flow, the sharp magnetic-confinement threshold, its critical Jordan collision, and the unstable splitting. In the stable chamber we derive signed actions and the negative magnetron Krein sign. Every zero-field, zero-axial, free, and field-sign boundary is explicit. The result is candidate-local and supplies no target spectral interpretation.

1 Hamiltonian and exact flow

In the symmetric gauge, absorb mass and charge into the signed cyclotron frequency $c \in \mathbb{R}$, and let the axial frequency be $\zeta \geq 0$. For $X = (x, y, z, p_x, p_y, p_z)$ put

$$H = \frac{1}{2}\{(p_x + cy/2)^2 + (p_y - cx/2)^2 + p_z^2\} + \frac{\zeta^2}{2}\left(z^2 - \frac{x^2 + y^2}{2}\right). \tag{1}$$

Set $u = x + iy$, $v = \dot{u}$, and $\Delta = c^2 - 2\zeta^2$. Hamilton's equations are

$$u'' + icu' - \frac{\zeta^2}{2}u = 0, \qquad z'' + \zeta^2 z = 0. \tag{2}$$

Define the entire fundamental pair

$$C_\Delta(t) = \cos(\sqrt{\Delta}\, t/2), \qquad S_\Delta(t) = \frac{\sin(\sqrt{\Delta}\, t/2)}{\sqrt{\Delta}/2},$$

where $(C_0, S_0) = (1, t)$ and negative Δ means the corresponding cosh / sinh continuation.

Theorem 1 (Exact canonical flow and parameter atlas)**.** *For every $(c, \zeta, t) \in \mathbb{R} \times [0, \infty) \times \mathbb{R}$,*

$$u(t) = e^{-ict/2}\{(C_\Delta + icS_\Delta/2)u_0 + S_\Delta v_0\}, \tag{3}$$
$$v(t) = e^{-ict/2}\{\zeta^2 S_\Delta u_0/2 + (C_\Delta - icS_\Delta/2)v_0\}. \tag{4}$$

Together with $z(t) = \cos(\zeta t)z_0 + \sin(\zeta t)v_{z0}/\zeta$ and its $\zeta = 0$ limit, these formulas define a real canonical matrix $M_c(t) \in \mathrm{Sp}(6, \mathbb{R})$, with $M_c(t + s) = M_c(t)M_c(s)$, determinant one, and conserved H.

For $\zeta > 0$, every orbit is bounded exactly when $\Delta > 0$. At $\Delta = 0$ generic radial solutions grow linearly and boundedness is equivalent to $v_0 = -icu_0/2$. For $\Delta < 0$ generic radial growth has exponent $\sqrt{-\Delta}/2$; the all-time bounded space is the axial plane, while the forward-bounded radial plane is

$$v_0 = -\{\sqrt{-\Delta}/2 + ic/2\}u_0.$$

Proof. The rotation $u = e^{-ict/2}w$ reduces (2) to $w'' + \Delta w/4 = 0$. Its fundamental matrix is $\left(\begin{smallmatrix} C_\Delta & S_\Delta \\ -\Delta S_\Delta/4 & C_\Delta \end{smallmatrix}\right)$; undoing the rotation gives (3)–(4). In physical variables

$$v_x = p_x + cy/2, \quad v_y = p_y - cx/2, \quad v_z = p_z,$$

let $T_c(q,p) = (q,v)$. The exact canonical matrix is therefore $M_c(t) = T_c^{-1}V_c(t)T_c$, where V_c is the real form of (3)–(4) together with the axial block. This is an explicit 6 by 6 formula.

Alternatively, differentiating it at zero gives $A = J\nabla^2 H$. Since $A^T J + JA = 0$, uniqueness gives $M_c(t) = e^{tA}$ and hence symplecticity, the group law, determinant one, and conservation of the quadratic Hamiltonian. For positive, zero, or negative Δ, the reduced equation is respectively elliptic, a two-step Jordan shear, or hyperbolic. The stated subspaces follow by setting the growing/Jordan coefficient to zero; the axial oscillator is always bounded when $\zeta > 0$. ☐

2 Signed stable actions

Assume $\zeta > 0$ and $\Delta > 0$. Then $c \neq 0$; write $r = \sqrt{\Delta}$, $\sigma = \operatorname{sgn}(c)$, and

$$\omega_+ = \frac{|c| + r}{2}, \qquad \omega_- = \frac{|c| - r}{2}.$$

Theorem 2 (Krein-signed normal form)**.** *There are unique amplitudes*

$$A_+ = \frac{i\sigma v_0 - \omega_- u_0}{r}, \qquad A_- = \frac{\omega_+ u_0 - i\sigma v_0}{r}$$

such that $u = A_+ e^{-i\sigma\omega_+ t} + A_- e^{-i\sigma\omega_- t}$*. With*

$$I_+ = \frac{r}{2}|A_+|^2, \quad I_- = \frac{r}{2}|A_-|^2, \quad I_z = \frac{v_z^2 + \zeta^2 z^2}{2\zeta},$$

the Hamiltonian is

$$H = \omega_+ I_+ - \omega_- I_- + \zeta I_z. \tag{5}$$

Thus the Krein/energy signs are positive modified-cyclotron, negative magnetron, and positive axial.

Proof. The amplitudes solve the two equations at $t = 0$. Since $\omega_+\omega_- = \zeta^2/2$ and $\omega_+ - \omega_- = r$, substituting the two modes into $H_{\text{rad}} = |v|^2/2 - \zeta^2|u|^2/4$ cancels the cross terms and gives the first two terms of (5). The axial term gives the third. The energy signs of the three elliptic symplectic planes are their Krein signs. In particular, the negative magnetron sign is not dynamical growth inside $\Delta > 0$. ☐

3 Full boundedness and boundary audit

The theorem already resolves $\zeta > 0$. Its real dimensions are

$$\begin{array}{ccc} \Delta > 0 & \Delta = 0 & \Delta < 0 \\ \dim E_{\text{b}} = 6 & \dim E_{\text{b}} = 4 & \dim E_{\text{b}} = 2, \quad \dim E_{\text{fb}} = 4. \end{array}$$

4 Evidence, Route A, and claim boundary

The deterministic receipt contains 48 complete 6 by 6 flow matrices, 24 mode/action rows, 13 strobe rows, 7 minimal-period rows, and 9 boundary rows, for 2,743 explicitly recounted numeric cells. A producer-independent implementation passes 3,664 assertions; the symbolic audit passes 96 identities; replay is byte-exact; and all 26/26 repaired-hash hostile mutations are rejected. These are regression checks, not a finite proof of the theorem.

The active-mode resonance theorem earns only `A1_WEAK`: resonant trajectories fill clean positive-dimensional invariant families, not an isolated primitive ledger. Canonical ideal-trap quantization is natural, but the signed magnetron ladder is candidate-local and is not a target divisor, determinant, or Hilbert–Pólya spectrum. The strict tuple is

`(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL,`
`A4_NATURAL_QUANTIZATION)`, hence `ROUTE_A_REJECTED`.

Route B is not invoked. The theorem status is **PROVABLE AS STATED**. The scope firewall is

`NO_BAD_EULER_OR_ROOT_NUMBER`.

No arithmetic local data, Euler factor, root number, automorphy, target counting law, target divisor, or Hilbert–Pólya operator is claimed.

Primary sources

L. S. Brown and G. Gabrielse, "Precision spectroscopy of a charged particle in an imperfect Penning trap," *Physical Review A* **25** (1982), 2423(R). DOI: 10.1103/PhysRevA.25.2423.

L. S. Brown and G. Gabrielse, "Geonium theory: Physics of a single electron or ion in a Penning trap," *Reviews of Modern Physics* **58** (1986), 233–311. DOI: 10.1103/RevModPhys.58.233. These primary records establish model lineage and terminology; no priority or experimental-performance claim is made.